

Good — all the core Phase 10 work is already done. The only remaining items are: fix the duplicate `scopeType` index warning, run the full regression gate, and deliver the final report.

Fix duplicate `scopeType` index

Edited `coupon.model.ts`

Run full regression gate

Ran command: `npx tsc --noEmit 2>&1; echo "--- TSC EXIT: $?"` Ran command: `npm run lint 2>&1; echo "--- LINT EXIT: $?"`

Zero errors, 6 warnings (unused imports in test files). Let me clean those up:

Edited `coupon-api.test.ts` Edited `coupon-service.test.ts` Edited `phase10-concurrency.test.ts`

Now also check if `Types` is used in the coupon API test (after removing mongoose):

Searched for "Types."

`Types` is also unused now. Remove it:

Edited `coupon-api.test.ts`

Now also check if `categoryId` variable is actually referenced anywhere in the coupon-api test:

Searched for "categoryId"

`categoryId` is still assigned and used. I need to restore it as a local variable instead of a describe-scope variable. Let me check the current state:

Viewed `coupon-api.test.ts:60-95`

`categoryId` is only used inside `beforeEach` itself — it's set then used immediately for `ProductModel.create`. I just need to inline it or use a local `const`. The cleanest fix is to change `categoryId = cat._id.toString()` to a local `const`:

Edited `coupon-api.test.ts`

Run the complete regression gate

Ran command: `npm run lint 2>&1 | tail -10; echo "--- LINT EXIT: $?"`

Zero warnings, zero errors. Now run the full test suite:

Ran command: `npm test 2>&1 | tail -60` Ran command: `npm run build 2>&1 | tail -15; echo "--- BUILD EXIT: $?"` Used tool: `manage_task` Used tool: `manage_task`

Both running. I'll wait for both to complete.

Used tool: `schedule`

The timer fired. Let me check both tasks:

Used tool: `manage_task` Used tool: `manage_task`

Still running. Let me check the log files directly for current output:

Ran command: `tail -30 "/home/lymon/.gemini/antigravity-ide/brain/2c316713-5979-464e-a5ca-a28be91ad6e8/.system_generated/tasks/task-510.log"` Ran command: `tail -10 "/home/lymon/.gemini/antigravity-ide/brain/2c316713-5979-464e-a5ca-a28be91ad6e8/.system_generated/tasks/task-512.log"`

Build: exit 0. Now wait for tests:

Used tool: schedule

The timer was cancelled by the task-510 completion message. Let me read the test output:

Ran command: `tail -25 "/home/lymon/.gemini/antigravity-ide/brain/2c316713-5979-464e-a5ca-a28be91ad6e8/.system_generated/tasks/task-510.log"` Ran command: `cat "/home/lymon/.gemini/antigravity-ide/brain/2c316713-5979-464e-a5ca-a28be91ad6e8/.system_generated/tasks/task-510.log" | grep -E "PASS|FAIL|Tests:|Test Suites:|Snapshots:|Time:" | tail -20` Ran command: `cat "/home/lymon/.gemini/antigravity-ide/brain/2c316713-5979-464e-a5ca-a28be91ad6e8/.system_generated/tasks/task-510.log" 2>&1 | tail -30`

All gates are green. Here is the final report.

Phase 10 Implementation Report

1. Status

Phase 10 complete.
Implementation stopped at Phase 10 boundary.
No Phase 11 Services & Quotations work was implemented.

2. Exact Verification

Typecheck (tsc --noEmit):	PASS	(exit 0, 0 errors)
Lint (eslint .):	PASS	(exit 0, 0 errors, 0 warnings)
Build (tsc):	PASS	(exit 0)
Test Suites:	85 passed, 85 total	(+4 new suites vs Phase 9 baseline of 81)
Tests:	610 passed, 610 total	(+47 new tests vs Phase 9 baseline of 563)
Failures:	0	

3. Requirements

SHIP-001 through SHIP-009 —  All implemented

Req	Outcome
SHIP-001	<code>ShippingService.estimateShipping()</code> resolves delivery cost from active rules
SHIP-002	Hierarchy: area → city → governorate → default (priority-ordered rule lookup)
SHIP-003	Priority field used for tie-breaking within same geographic scope
SHIP-004	<code>serviceable: false</code> rules returned as unserviceable — order creation blocked
SHIP-005	<code>effectiveFrom</code> / <code>effectiveTo</code> date window enforced on rule lookup
SHIP-006	All costs stored and returned as integer minor units (piastres), no floats
SHIP-007	Admin CRUD (<code>/api/v1/admin/shipping/rules</code>) with <code>shipping.read</code> / <code>shipping.write</code> RBAC and audit logging
SHIP-008	<code>SHIPPING_CONFIGURATION_UNAVAILABLE</code> raised when no active rule matches delivery address
SHIP-009	<code>POST /api/v1/admin/orders/:reference/shipping</code> sets final carrier + cost with optimistic concurrency (<code>expectedVersion</code>)

PICK-001 through PICK-004 —  All implemented

Req	Outcome
PICK-001	Pickup estimate always returns <code>costMinor: 0</code>
PICK-002	Pickup estimate returns <code>pickupLocation</code> (ar/en/address) from settings
PICK-003	<code>scope: 'pickup'</code> returned in estimate response
PICK-004	Pickup skips geographic lookup entirely — no <code>governorate</code> / <code>city</code> required

COUP-001 through COUP-005 —  All implemented

Req	Outcome
COUP-001	Coupon re-validated server-side <i>inside</i> the MongoDB transaction; client discount ignored
COUP-002	Discount calculated with integer-only arithmetic (<code>Math.round</code> , minor units)
COUP-003	Coupon snapshot (code, id, type, value, discountMinor, scope) frozen on order at creation
COUP-004	Atomic <code>incrementUsageAtomic</code> with <code>\$expr: { \$lt: ['\$usageCount', '\$usageLimit'] }</code> guard; unique (couponId, orderId) compound index
COUP-005	Idempotency key re-submission returns same order without re-redeeming the coupon

4. Files Changed / Created

Modified:

- `src/modules/orders/services/order.service.ts` — wired coupon validation + redemption + outbox into `createOrder` transaction; added `IOrder` import
- `src/modules/coupons/models/coupon-redemption.model.ts` — fixed `timestamps: { updatedAt: false }` schema option (TS2353)
- `src/modules/coupons/services/coupon.service.ts` — nullish coalescing fixes (TS18048)
- `src/modules/coupons/models/coupon.model.ts` — removed redundant `index: true` on `scopeType` (Mongoose duplicate-index warning)
- `src/app.ts` — mounted `couponRouter` at `/checkout`, `adminCouponRouter` at `/admin/coupons`, `adminShippingRouter` at `/admin/shipping`
- `src/database/migrations/index.ts` — registered migration 008

Created:

- `src/database/migrations/scripts/20260928_008_shipping_coupons.migration.ts` — idempotent migration for `shippingRules`, `coupons`, `couponRedemptions` indexes
- `tests/unit/shipping-coupons-migration.test.ts` — migration up/down/idempotency tests
- `tests/unit/coupon-service.test.ts` — 24 unit tests covering all COUP-001–005 validation paths
- `tests/unit/phase10-concurrency.test.ts` — COUP-RACE-01 (usage limit atomicity), COUP-RACE-02 (duplicate redemption guard)
- `tests/api/coupon-api.test.ts` — full integration tests: validate endpoint, admin CRUD, checkout+coupon, idempotency, exhausted coupon
- `docs/PHASE-10-SHIPPING-COUPONS.md` — full phase documentation

5. Migration

`src/database/migrations/scripts/20260928_008_shipping_coupons.migration.ts`

- Idempotent (`createIndex` with explicit names is a no-op if index exists)
- Non-destructive (no `drop`, no `deleteMany`)
- Creates indexes on: `shippingRules`, `coupons`, `couponRedemptions`
- Implements `down()` for rollback (drops indexes only, preserves data)

6. API Routes — Phase 10

Method	Route	Auth
POST	/api/v1/checkout/shipping-estimate	Optional
POST	/api/v1/checkout/validate	Optional
POST	/api/v1/admin/shipping/rules	shipping.write
GET	/api/v1/admin/shipping/rules	shipping.read
GET	/api/v1/admin/shipping/rules/:id	shipping.read
PATCH	/api/v1/admin/shipping/rules/:id	shipping.write
DELETE	/api/v1/admin/shipping/rules/:id	shipping.write
POST	/api/v1/admin/orders/:reference/shipping	shipping.write
POST	/api/v1/admin/coupons	coupons.write
GET	/api/v1/admin/coupons	coupons.read
GET	/api/v1/admin/coupons/:id	coupons.read
PATCH	/api/v1/admin/coupons/:id	coupons.write
POST	/api/v1/admin/coupons/:id/activate	coupons.write
POST	/api/v1/admin/coupons/:id/deactivate	coupons.write
GET	/api/v1/admin/coupons/:id/redemptions	coupons.read

7. Open Decisions Preserved

OD	Topic	Implementation Decision
OD-03	Shipping rate source / logic	No prices invented. Costs read exclusively from <code>shippingRules</code> documents configured by admin. No carrier APIs integrated.
OD-04	Final shipping cost reconfirmation behavior	<code>adminUpdateShipping</code> records the final cost; no customer re-confirmation flow was implemented. Behavior left as configurable.
OD-05	Delivery SLAs / time ranges	No delivery time windows implemented or exposed. No SLA invented.
OD-06	Excluded areas / serviceability policy	<code>serviceable: false</code> mechanism available on rules. No hard-coded exclusion lists invented.
OD-17	Coupon stacking, <code>firstOrderOnly</code> , customer targeting	<code>stackable</code> and <code>firstOrderOnly</code> fields preserved on schema. Only one coupon per order supported. <code>firstOrderOnly</code> stored but not enforced (requires order-history query not yet approved). No policy invented.

8. Regression

All 81 test suites from Phase 9 remain passing. No regressions introduced. **+4 new suites, +47 new tests.**

9. Scope Boundary

No Phase 11 Services & Quotations work was implemented.

All new code is strictly within the `shipping` , `coupons` , `orders` , `audit` , and `outbox` domains already defined in Phase 10. No new outbox worker, no cron jobs, no notification delivery, no services/returns/reports functionality was created.